

Highway Security Reviews

Public Executive Summary | v2.1 package | September 29, 2026

Cyfrin reviewed the EVM, Substrate, and Solana/SVM implementations of Highway, a cross-chain token-and-message bridge. A source chain emits a message, a relayer committee attests to it, and a destination chain independently reconstructs and verifies it before executing a token transfer or payload.

This public summary describes the scope, approach, findings, and remediation status across the three reviews. The full technical reports remain controlling for exact file scope, trust assumptions, individual findings, Highway responses, remediation evidence, and Cyfrin verification notes.

Scope and review periods

Implementation	Audited repository and commit	Review period	Principal scope
EVM	Project-Highway/hway-ethereum 73def9a5bdc5a82aae6e5f691db4d7fd2b5c6198	June 8-26, 2026	Message entry/execution; committee and relayer registries; storage, escrow, fees, replay protection, payload controls, BLS12-381, proxies, and access control.
Substrate	Project-Highway/hway-substrate df99567ea9084d5dcf7e6eee3ab6cb3478c3c160	June 29-July 17, 2026	Highway entry, registry, configuration, BLS utilities, NFT delegation, permission, and relayer backing.
Solana/SVM	Project-Highway/hway-solana b8246bbd6d6866b31d539e565700815188150541	July 20-31, 2026	Configuration, outbound entry, two-step inbound execution, relayer registration, active-set management, message IDs, constants, and supporting deployment tooling.

Results and remediation status

The three v2.1 reports contain 161 findings. Of these, 121 are resolved, one is partially resolved, and 39 are acknowledged. Every Critical and High finding is resolved.

Severity	Total	Resolved	Partially resolved	Acknowledged
Critical	2	2	0	0
High	2	2	0	0
Medium	15	9	0	6
Low	90	65	1	24
Informational	46	37	0	9
Gas optimization	6	6	0	0
Total	161	121	1	39

EVM 42 27 resolved 15 acknowledged	Solana/SVM 80 69 resolved 11 acknowledged	Substrate 39 25 resolved 1 partial 13 acknowledged
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How to read the statuses

Resolved means the corresponding report records the issue as resolved with a Cyfrin verification note. Partially resolved applies only to Substrate L-30: one reported defect was fixed while the remaining documented behaviors were retained under the stated system invariant. Acknowledged means Highway chose to retain the behavior or accept the residual risk; it is not equivalent to resolution or endorsement by Cyfrin. The reports record Cyfrin's position under every acknowledged finding: 8 rationales accepted, 31 accepted with a condition, and 0 not accepted.

Review boundaries

The formal scope is the exact file list in each report. The reviews do not cover every repository file, every dependency, the complete off-chain relayer service, unreviewed changes, or an arbitrary live deployment. Production configuration was not independently verified. Runtime, deployment, and configuration values are relevant only in the context stated by each report, including where a value is fixed in code.

Method and effort

Cyfrin performed three manual security reviews over the periods above. The work examined architecture and trust assumptions; traced message and value flows; and reviewed committee selection, aggregate-signature verification, message encoding, replay protection, token custody, payload dispatch, administration, and configuration. Cross-implementation review covered the message-ID preimage, committee-attestation payload, epoch handling, domain separation, proof-of-possession behavior, committee bitmap interpretation, and chain-specific cryptographic differences. Submitted mitigations were reviewed against code and supporting evidence where the reports state that Cyfrin verified a resolution. Review windows are dates, not person-hour estimates.

Cross-implementation consistency

Highway depends on independently implemented legs reproducing compatible application-level message IDs, attestation preimages, and committee-selection rules. Current EVM and Substrate production source share the network-bound V1 message-ID layout and the 94-byte epoch-bound attestation preimage; Solana shares the V1 application preimages while using curve-specific BN254 verification. The checked-in Substrate helper and diagnostic fixture remain on a pre-epoch path, so this summary does not make an unconditional end-to-end release-parity claim beyond the code paths and evidence documented in the reports. This qualification does not mean the three production implementations are known to disagree: the reviewed production paths align on the application-level message and attestation formats, but the available evidence does not include one current end-to-end compatibility run covering all three implementations.

Remediation overview

Highway remediated the highest-severity defects and supplied changes for many lower-severity findings. Finding-level status and verification evidence in the full reports remain controlling. Residual risk is concentrated in acknowledged findings, the partially resolved Substrate item, privileged operational assumptions, cryptographic and key-lifecycle hardening, payload-integration boundaries, and continued cross-leg compatibility testing.

Implementation highlights

EVM

The EVM review identified one High, seven Medium, and lower-severity and gas findings. The resolved High issue concerned a non-injective message-ID encoding. Other themes included authenticated source context, cross-deployment encoding, committee and key lifecycle behavior, payload execution and recovery, BLS input validation, and privileged deployment and administration paths.

Substrate

The Substrate review identified one Critical, five Medium, and lower-severity findings. The resolved Critical issue concerned a mismatch between bitmap bits counted toward the signature threshold and committee keys aggregated for verification. Highway raised the relevant runtime parameters and added an out-of-window bitmap guard; Cyfrin verified that the audited threshold-bypass path was closed. Remaining themes include payload-dispatch origin, BLS registration, relay lifecycle, weight and value accounting, delegation, slashing, and configuration controls.

Solana/SVM

The Solana review identified one Critical, one High, three Medium, and lower-severity and gas findings. The resolved Critical issue concerned mapping publicly known scalars directly onto the curve in attestation and proof-of-possession verification. The resolved High issue concerned a native-token corridor that could accept outbound escrow without a compatible inbound settlement path. Other themes included committee rotation, vault validation, payload-account binding, cryptographic input handling, resource limits, access control, and observability.

Closing assessment

Highway remediated the highest-severity issues, and all Critical and High findings are recorded as resolved and verified. The reviews identified material security defects across all three implementations, including defects capable of undermining threshold verification or preventing correct settlement. This represents a material improvement over the audited revisions.

Residual risk remains. Coordinated releases and compatibility testing are important because each leg independently reconstructs authenticated messages and attestations. This assessment is limited to the documented scope, audited commits, and specifically reviewed mitigations. It is not a product rating, warranty, guarantee, or endorsement, and it does not establish the security of an arbitrary production deployment or later code revision.

Full reports

The report files distributed with this public package are identified below. Their SHA-256 digests allow readers to verify copies distributed or mirrored elsewhere.

Implementation	Report filename	PDF SHA-256
EVM	2026-09-29-cyfrin-highway-evm-v2.1.pdf	c89d2a996ca2edcaa74f96121bd8e1e3 7b234484e18fec0df183597db9d5bee5
Substrate	2026-09-29-cyfrin-highway-substrate-v2.1.pdf	e0de8b06cf07687ab7196904f4bf2f54 3e9a388e31c13aaf9113a24492cffc90
Solana/SVM	2026-09-29-cyfrin-highway-solana-v2.1.pdf	103c14f1d9f352a3b1160eb1d886120b 519e664c0eb6782ba1fd12b288ab3d41